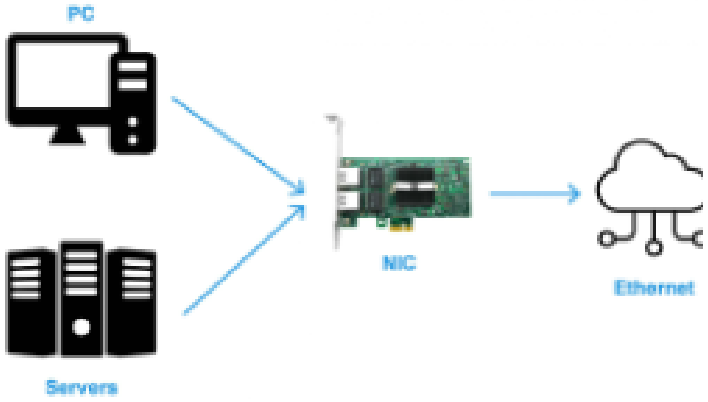
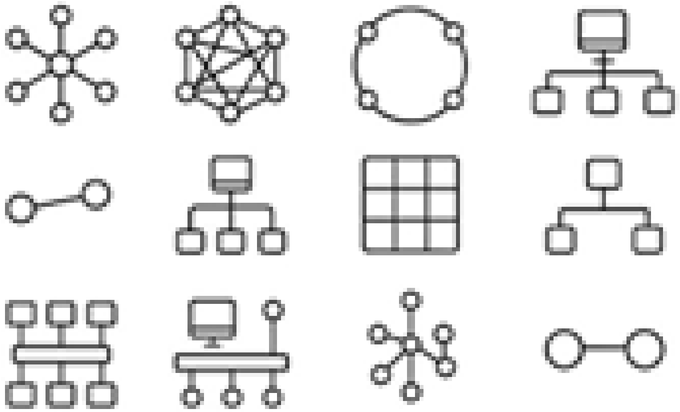
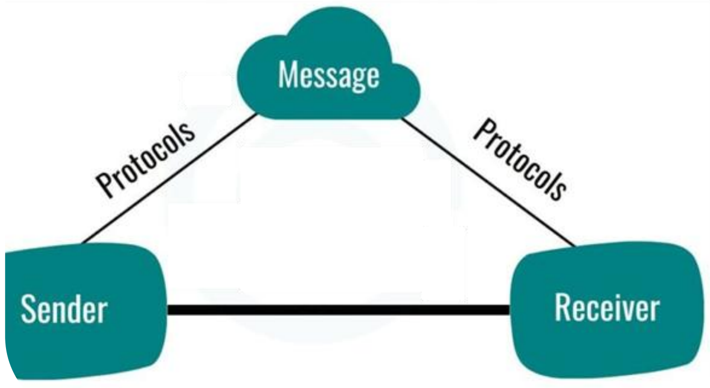
Key elements of IoT networking







Network Interface

Hardware or software component that connects a device to a network

Network Topologies

Arrangement of a network's components, and how data flows through them

Data Communication Protocols

the rules for how data is transmitted and received across the network

Key elements of IoT networking

Network Interfaces—Hardware or software components that connect an IoT device to a network. They're the gateway for communication.

Wired

- USB
- Ethernet
- RS232 & RS485

Wireless (short Distance)

- Bluetooth
- BLE
- ZigBee
- Z-Wave
- Wi-Fi

Wireless (Long Distance)

- LoRaWAN
- 2G/3G/4G
- 5G
- NB-IoT

Key elements of IoT networking

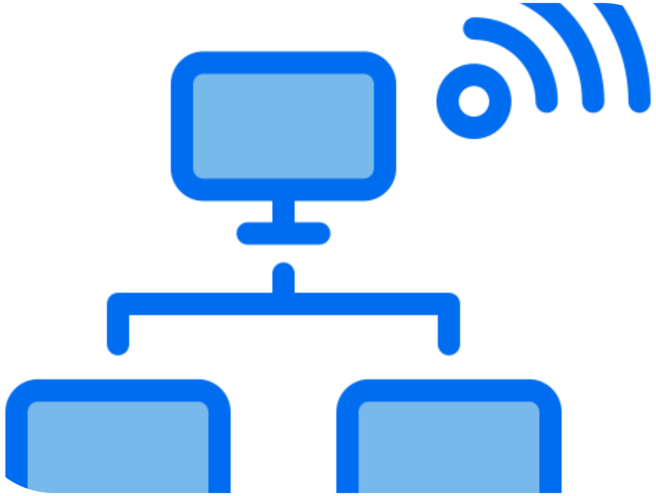
Network Topology – arrangement of a network's components, and how data flows through them.

Point-to-Point, or P2P

2 devices connected directly to each other. (RS232 or Bluetooth_ links.

Mesh Topology

devices are interconnected so that data can take multiple paths to reach its destination



Wireless LAN

star topology where all devices connect to a central access point—usually over Wi-Fi

Key elements of IoT networking

Data Communication Protocol – rules and standards that define how data is transmitted and received across a network

Traditional networking	Messaging & emails	Specialized for IoT
HTTP	SMTP	MQTT
FTP	XMPP	CoAP

Why multiple Protocols

Need for multiple protocol – to Balance cost and capability for various use cases

No one-size-fits-all protocol for IoT

Purpose-made Protocols for IoT are seeing increased adoption

Software is becoming more important to managing IoT connectivity

"Ease of use" and "reliability" are most important when choosing a new protocol

Decision-making for IoT protocols has become a multi-stakeholder exercise

Key Factors to be considered

Cost

Cost : N/w stack processing cost, transceiver cost based on distance

Bandwidth – optimal bandwidth based on network topology , distance, peak bandwidth required in burst mode

Power Usage– Power consumption in processing , data transfer (wired/wireless)

Infrastructural Attributes: Security, latency, simultaneous connections